Fundamentals of Programming (CS-114)

Numeric Data Types and Printing of Data and Data Types

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Topic 3 - Numeric Data Types and Printing of Data and Data Types

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RECAP

- Python is a free programming language licensed under PSF which can be downloaded, distributed, installed, and used free (libre).
- Python packages are released by PyPI.
- Anaconda provides python with several packages incorporated which is very useful specially for the newbies.
- Python has different working environments including interactive mode, text and terminal, and IDE arrangements.
- The installer provided by Anaconda and Spyder IDE (because it has MATLAB like layout as well) are highly recommended for scientific computing.

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OBJECTIVES

After taking this lecture and studying, you should be able to

- Describe in your own words different data types in human communication.
- Describe different data types in python in general and numeric data types in particular in python, in your own words.
- Use `print` function to display data types and data itself.

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THE WHY SECTION

- In objectives section, you encountered the term data types.
- In our daily life, we speak, read, write, and listen.
- In speaking, reading, writing, or listening, do we have data types?
- If yes, what data types we use in speaking, reading, writing, or listening?
- To what physical attributes, the data types are linked (reason of data types)?
- Does python programming (in fact all computer programming languages) also have the same data types?
- Are there additional data types in python? If yes, which additional data types python programming posseses and what are the reasons of having additional data types?

THE WHY SECTION

- Clinical digital thermomoeters sense the body temperature of a patient.
- They also display the temperature on the basis of which docotors take actions.
- It means displaying data in some cases is very important.

Figure. Clinical digital thermometer displaying temperature on the display.

THE WHY SECTION

- When we perform computations, we use data and the outcome of computations is also data.
- The output data need to be displayed for authorities to take action on the basis of outcome of the computation.
- Does python programming has the ability to display data on the standard output display device (LCD display)?
- If yes, how to display data on standard output display device.
- We answer to all these questions shortly.

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DATA TYPES

- Once I decided to bring the soft drink category ice cream soda, sold under the brand name of Pakola.
- I decided to purchase twelve (12) cans plus a two and a quarter of a liter (2.25 L) of the drink.

Figure. Pakola ice cream soda twelve cans (left) plus two and a quarter of a liter (2.2 L) (right).

DATA TYPES

- I went to the store and said, "Give me 12 cans and a 2.25 L of Pakola ice cream soda."
- In the above quote statement, 12 and 2.25 are the numbers and everything else is a text data, you already know.
- You also know that 12 is a whole number and 2.25 is a decimal point number.
- My question is, "To what attributes the whole numbers and decimal point numbers, in general, are linked?"
- Let's start by breaking the flow of the lecture (in fact the lecture is in flow), if you count the number of cars in the car parking, can you say 20.25 cars or 20.93 cars?

DATA TYPES

- Definitely, no.
- You will tell a whole number like 19 or 23 etc.
- Now, another question, "Is there any fractional error in the number of cars, you counted in the parking?"
- Definitely, no.
- It means, when we count objects, we use whole numbers and there is no fractional error in the whole numbers.

DATA TYPES

Whole Numbers

Whole numbers are attributed to counting of objects with no fractional error just like in counting the number of cars in the car parking or asking to give 12 cans of ice cream soda to the seller in the store. In computer programming, whole numbers are termed as integers represented by `int`.

- So, just like real world, when we have to count in programming, we use `int` data type.
- Suppose, you are performing an experiment in the lab and you measure the mass of sodium chloride.
- You say the mass of the salt is 2.5 grams.
- Can you claim that the salt is exactly (100%) 2.5 grams?

DATA TYPES

- Definitely, no.
- Let's play with very simple computations of $\left(\frac{1}{50}\right) \times 50$ and $\left(\frac{1}{49}\right) \times 49$.
- We expect $\left(\frac{1}{50}\right) \times 50 = 1.0$ and $\left(\frac{1}{49}\right) \times 49 = 1.0$
- But, computers don't give the output as expected.
- The answer lies in the output of 1) $\frac{1}{49}$ and $\frac{1}{50}$ and 2) displaying the results to at least 19 places of decimal.
- As we see, the fractional numbers are accurate upto 18 places of decimal in python and the 19th place has error.
- In the real physical world also, fractional numbers are not 100% certain.

DATA TYPES

- It means, when we measure things, we use fractional numbers which always have colorgreenerror.

Fractional Numbers

Fractional numbers are attributed to the measurements which always have error just like in the measurement of sodium chloride in the lab or the measurement of volume of ice cream soda filled in a bottle by the company. In computer programming, fractional data is termed as floating point numbers represented by `float`.

- Any kind of english (or other) text is called the text data type in real world whereas in python programming it is termed as string data type represented by `str` .
- The `int` and `float` data types fall under the category of numbers on which arithmetic operations are possible whereas impossible on `str` data type.

DATA TYPES

Python Data Types

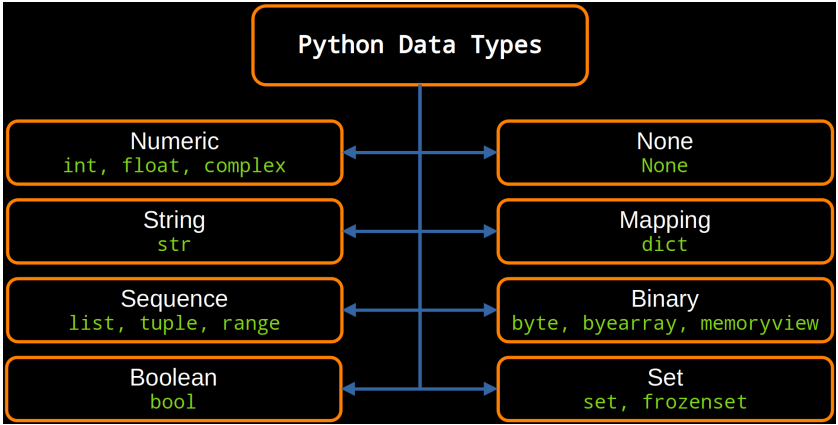

Figure. Data types in python programming.

DATA TYPES

- In the figure shown above, we can see that only three data subtypes are in use in human communication which are `int`, `float`, and `str`.
- Python has additional data types because these data types belong to science and engineering (all humans are not scientists and engineers) and to work with science, additional data types are required.
- In the same way, human also have several data types which cannot be used in computers *e.g.*, communication in symbolic form etc.
- In other words, data types are domain specific and therefore human communication, animal communication, and computers have different data types of their own domain.
- In the data types shown, we will touch numeric, string, sequence, boolean, and none data tpes as they are of our use and others will be rarely used.

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NUMERIC DATA TYPES

- Numeric data types are of three types.

  1. Integers
  2. Floating-Point Numbers
  3. Complex

- Now, we cover these data types one by one.
- These data types are the most used data types in science and a solid understanding will help write good computer programs.
- They are not only important in science and engineering, but also in all other domains of work.

NUMERIC DATA TYPES

1. Integers

- Integers are the whole numbers.
- They are attributed to counting of objects.
- Arithmetic operations are applicable and exact.
- Examples of integers are 29, -40, and 11 etc.
- Thousand, million, and billion separation are application with the use of underscore symbol *e.g.*, 299,792,458 can be written as `299_792_458` in python.
- The python returns `int` data type for integers.

NUMERIC DATA TYPES

2. Floating-Point Numbers

- Floating-point numbers are decimal point numbers.
- They are attributed to measurements.
- Arithmetic operations are applicable and accurate to 18 places of decimal in python.
- Examples of integers are 3.14159, 8.314, and -2.00233184123 (electron g-factor) etc.
- They can also be represented in scientific notation *e.g.*, $6.62607015 \times 10^{-34}$ and 6.67430×10^{-11} etc.
- The python returns `float` data type for integers.

NUMERIC DATA TYPES

3. Complex

- Complex data type consists of two part namely real part and imaginary part.
- They are attributed to complex numbers.
- Arithmetic operations are applicable as per mathematics rules for complex numbers.
- Examples of complex numbers are $3 + 2i$ and $3 - 2i$ etc.
- In python, they are represented as `3+2j`, `3-2j`, and `-4j` etc.
- The python returns `complex` data type for complex numbers.

NUMERIC DATA TYPES

- The real part of a complex number can be called by the following line of code in python.

```
>>>(3.2-1.5j).real
```

- The imaginary part of a complex number can be called by the following line of code in python.

```
>>>(3.2-1.5j).imag
```

Caution!

The symbol `>>>` is known as python prompt and you don't need to type it. Always type the text after the python prompt symbol, for example, `3+2j` only.

NUMERIC DATA TYPES

- We know that complex numbers always exist in conjugate pairs.
- Python has the facility to generate the conjugate pair of the given complex number automatically.
- The conjugate pair can be called by the following line of code on python prompt.

```
>>>(3-1.2j).conjugate()
```

Methods and Attributes

In calling real and imaginary parts, we used the keywords `real` and `imag` (no paranthesis after keyword), however, we used `conjugate()` keyword (paranthesis after keyword). The keywords `real` and `imag` are called the attributes while the keyword `conjugate()` is called a method. We will learn about these two terms in the upcoming lectures.

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PRINTING DATA AND DATA TYPES

- We already discussed that sometimes it is important to dispaly the value of a data type for the user.
- This can be achieved using `print` function in python.
- In python any keyword and following paranthesis is knows as the function, for example, `print` functions prints the value of the data type provided within paranthesis.
- We will look into functions in detail in the upcoming lectures.
- For now, we print the values of different data types and see the types of the data as well.

PRINTING DATA AND DATA TYPES

- Python shows Avogadro's number 6.022×10^{23} a `float`.
- Is Avogadro's number 6.022×10^{23} an `int` or `float`?
- Polling.
- Clarification.

PRINTING DATA AND DATA TYPES

- Python shows Avogadro's number 6.022×10^{23} a float.
- Is Avogadro's number 6.022×10^{23} an `int` or `float`?
- Polling.
- Clarification.

Caution!

In any case, human logic always prevails. Before diving into computer programming, you must be very clear about the scientific background of the topic of interest. Otherwise, confusions and misconceptions will arise and sometimes, results might also be misleading and wrong.

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DATA TYPE CONVERSION

- The functions `int`, `float`, and `complex` are used to convert a given data type into integers, floating-point, and complex numbers, respectively.
- However, convrsion depends on the input data type to function used.
- Following table clarifies it.

Data Type Conversion

Input Data Type	**Conversion To**		
	`int`	`float`	`complex`
`int`	Yes	Yes	Yes
`float`	Yes	Yes	Yes
`complex`	No	No	Yes

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SUMMARY

- In human communication, there are three major data types which are whole numbers, decimal point numbers, and text.
- In python programming there are eight major categories of data types and each of them has subtype(s).
- Numeric data types in python are `int`, `float`, and `complex`.
- The data is printed by using `print` function.
- The type of a data can be found by the `type` function.
- You must have a sound knowledge of the topic on which some computer program is required.

THANK YOU